

サンプルシステム2 データベース(SQLite3) 作成手順

2023/7/3

アーキテクトサービスセンター

堀 扶



目次

- インストール手順
- データベース作成/基本操作手順
- アンインストール手順



SQLiteサイトのDownloadを選択

The screenshot shows the SQLite Home Page in a web browser. The browser's address bar displays 'sqlite.org/index.html'. The page features the SQLite logo and the tagline 'Small. Fast. Reliable. Choose any three.' A navigation menu at the top includes links for Home, About, Documentation, Download, License, Support, and Purchase. The 'Download' link is highlighted with a red rectangular box. Below the navigation menu, the 'What Is SQLite?' section provides an overview of the database engine. To the right, a 'Common Links' sidebar lists various resources. At the bottom, the 'Latest Release' section shows the current version and download links. The footer mentions the SQLite Consortium members, including Bloomberg Engineering, Expensify, Navigation Data Standard, and Bentley.

SQLite Home Page

sqlite.org/index.html

SQLite

Small. Fast. Reliable.
Choose any three.

Home About Documentation **Download** License Support Purchase Search

What Is SQLite?

SQLite is a C-language library that implements a [small](#), [fast](#), [self-contained](#), [high-reliability](#), [full-featured](#), SQL database engine. SQLite is the [most used](#) database engine in the world. SQLite is built into all mobile phones and most computers and comes bundled inside countless other applications that people use every day. [More Information...](#)

The SQLite [file format](#) is stable, cross-platform, and backwards compatible and the developers pledge to keep it that way [through the year 2050](#). SQLite database files are commonly used as containers to transfer rich content between systems [\[1\]](#) [\[2\]](#) [\[3\]](#) and as a long-term archival format for data [\[4\]](#). There are over 1 trillion (1e12) SQLite databases in active use [\[5\]](#).

SQLite [source code](#) is in the [public-domain](#) and is free to everyone to use for any purpose.

Latest Release

[Version 3.42.0](#) (2023-05-16). [Download](#) [Prior Releases](#)

Common Links

- Features
- When to use SQLite
- Getting Started
- Try it live!
- Prior Releases
- SQL Syntax
 - Pragma
 - SQL functions
 - Date & time functions
 - Aggregate functions
 - Window functions
 - Math functions
 - JSON functions
- C/C++ Interface Spec
 - Introduction
 - List of C-language APIs
- The TCL Interface Spec
- Quirks and Gotchas
- Frequently Asked Questions
- Commit History
- Bugs
- News

Ongoing development and support of SQLite is made possible in part by [SQLite Consortium](#) members, including:

Bloomberg
Engineering

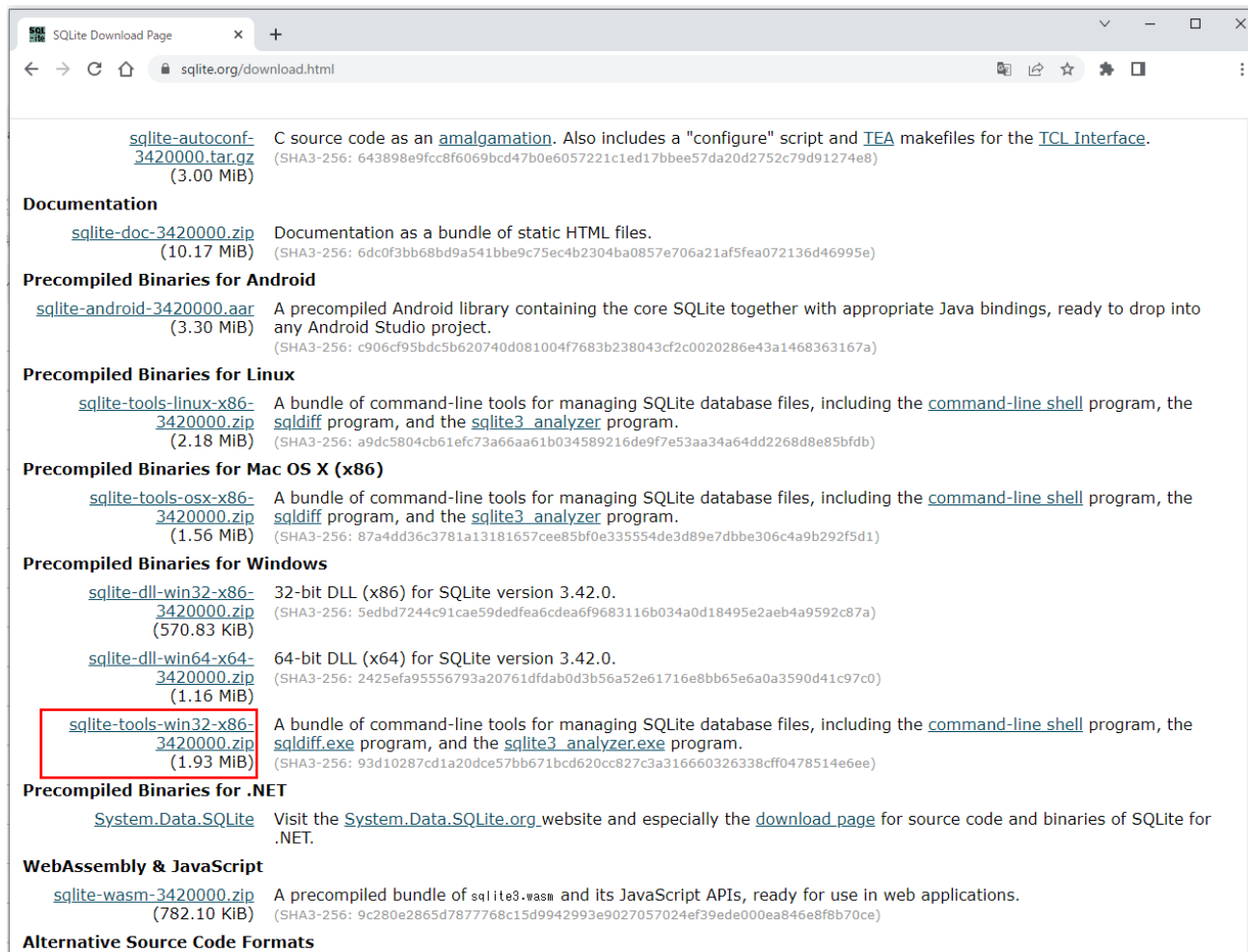
Expensify

Navigation Data Standard

Bentley
Sustaining Infrastructure



Precompiled Binariesを選択



SQLite Download Page

[sqlite-autoconf-3420000.tar.gz](#) (3.00 MiB) C source code as an [amalgamation](#). Also includes a "configure" script and [TEA](#) makefiles for the [TCL Interface](#). (SHA3-256: 643898e9fcc8f6069bcd47b0e6057221c1ed17bbe57da20d2752c79d91274e8)

Documentation

[sqlite-doc-3420000.zip](#) (10.17 MiB) Documentation as a bundle of static HTML files. (SHA3-256: 6dc0f3bb68bd9a541bbe9c75ec4b2304ba0857e706a21af5ea072136d46995e)

Precompiled Binaries for Android

[sqlite-android-3420000.aar](#) (3.30 MiB) A precompiled Android library containing the core SQLite together with appropriate Java bindings, ready to drop into any Android Studio project. (SHA3-256: c906cf95bdc5b620740d081004f7683b238043cf2c0020286e43a1468363167a)

Precompiled Binaries for Linux

[sqlite-tools-linux-x86-3420000.zip](#) (2.18 MiB) A bundle of command-line tools for managing SQLite database files, including the [command-line shell](#) program, the [sqldiff](#) program, and the [sqlite3_analyzer](#) program. (SHA3-256: a9dc5804cb61efc73a66aa61b034589216de9f7e53aa34a64dd2268d8e85bfdb)

Precompiled Binaries for Mac OS X (x86)

[sqlite-tools-osx-x86-3420000.zip](#) (1.56 MiB) A bundle of command-line tools for managing SQLite database files, including the [command-line shell](#) program, the [sqldiff](#) program, and the [sqlite3_analyzer](#) program. (SHA3-256: 87a4dd36c3781a13181657cee85bf0e335554de3d89e7dbbe306c4a9b292f5d1)

Precompiled Binaries for Windows

[sqlite-dll-win32-x86-3420000.zip](#) (570.83 KiB) 32-bit DLL (x86) for SQLite version 3.42.0. (SHA3-256: 5edbd7244c91cae59dedfea6cdea6f9683116b034a0d18495e2aeb4a9592c87a)

[sqlite-dll-win64-x64-3420000.zip](#) (1.16 MiB) 64-bit DLL (x64) for SQLite version 3.42.0. (SHA3-256: 2425efa95556793a20761dfdab0d3b56a52e61716e8bb65e6a0a3590d41c97c0)

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Precompiled Binaries for .NET

[System.Data.SQLite](#) Visit the [System.Data.SQLite.org](#) website and especially the [download page](#) for source code and binaries of SQLite for .NET.

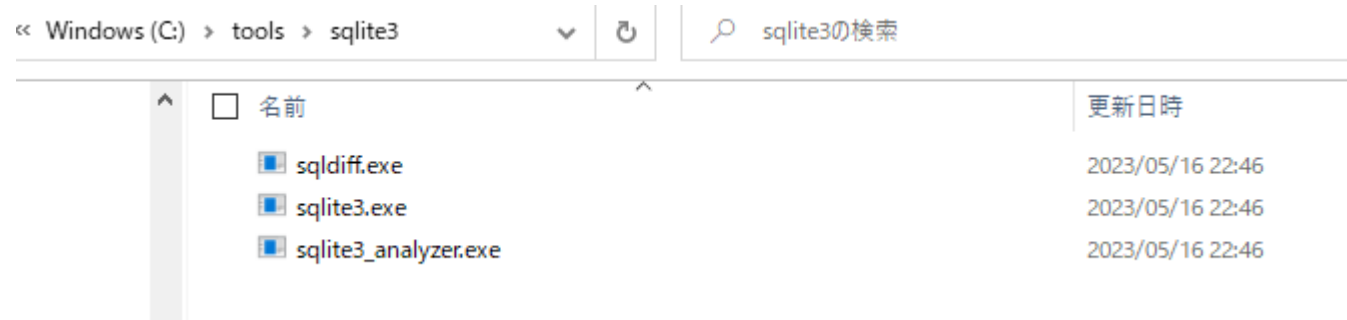
WebAssembly & JavaScript

[sqlite-wasm-3420000.zip](#) (782.10 KiB) A precompiled bundle of `sqlite3.wasm` and its JavaScript APIs, ready for use in web applications. (SHA3-256: 9c280e2865d7877768c15d9942993e9027057024ef39ede000ea846e8f8b70ce)

Alternative Source Code Formats



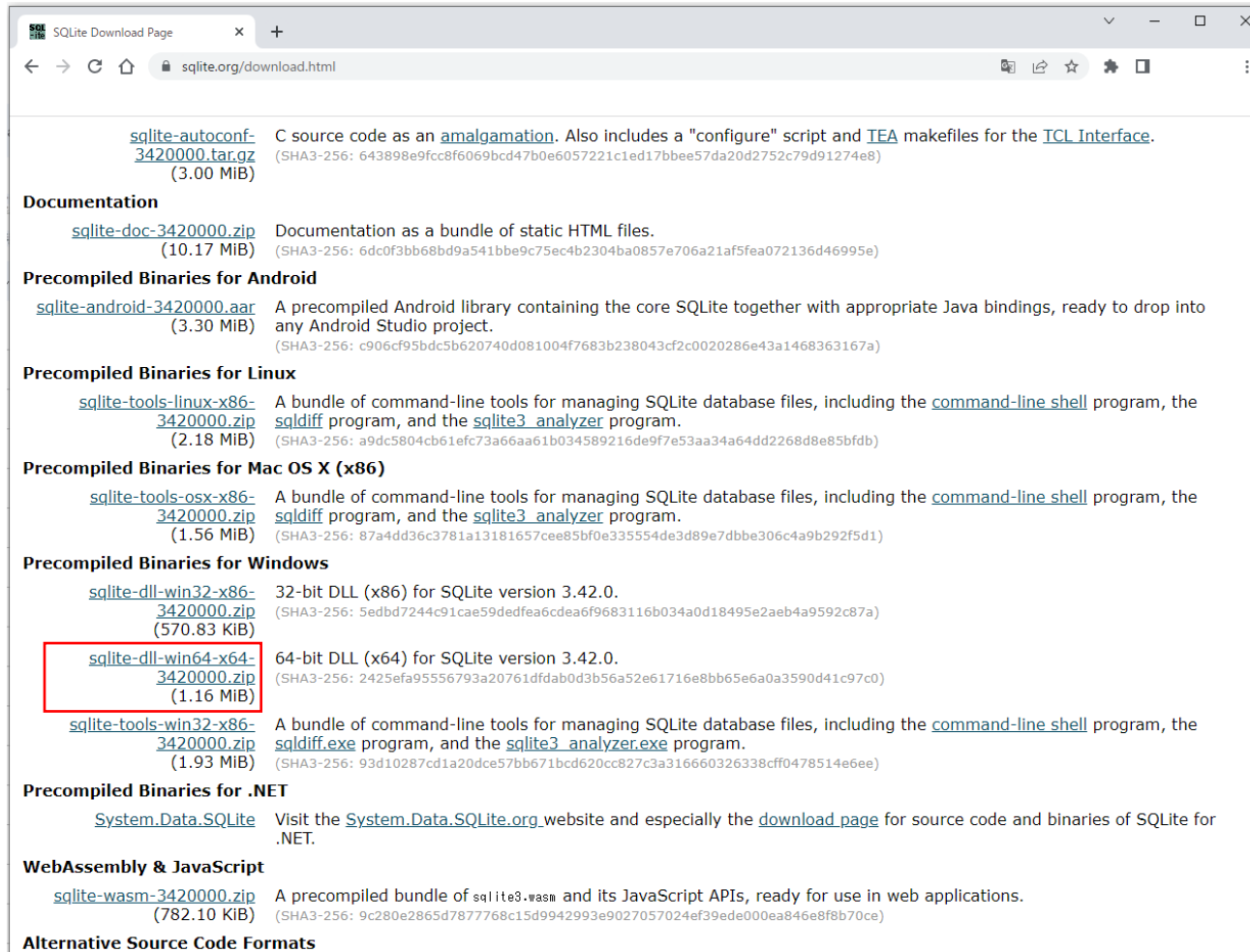
解凍して適当なフォルダへ移動



本資料では以下のフォルダへ移動した前提とする
C:¥tools¥sqlite3



DLLも選択



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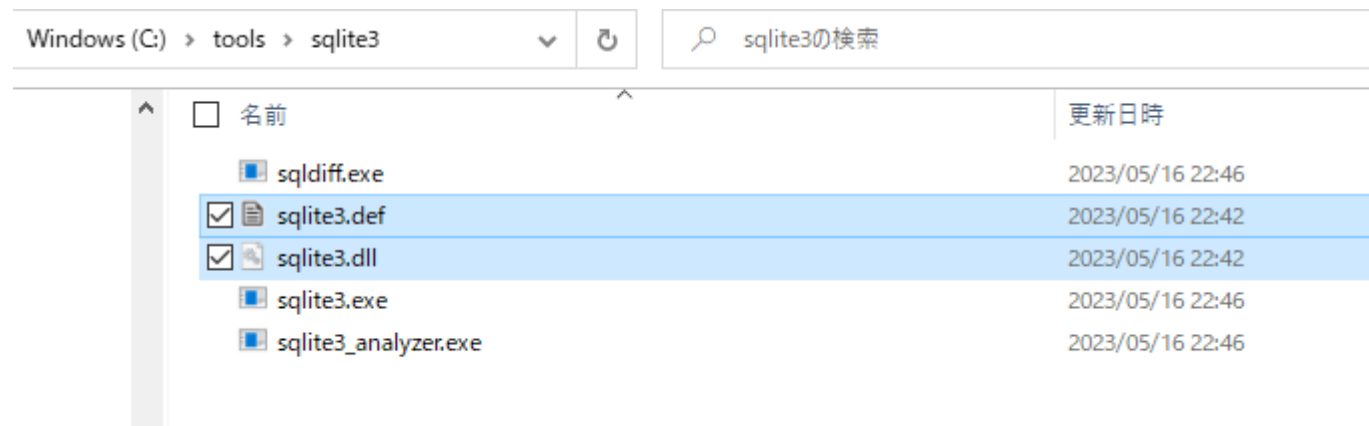
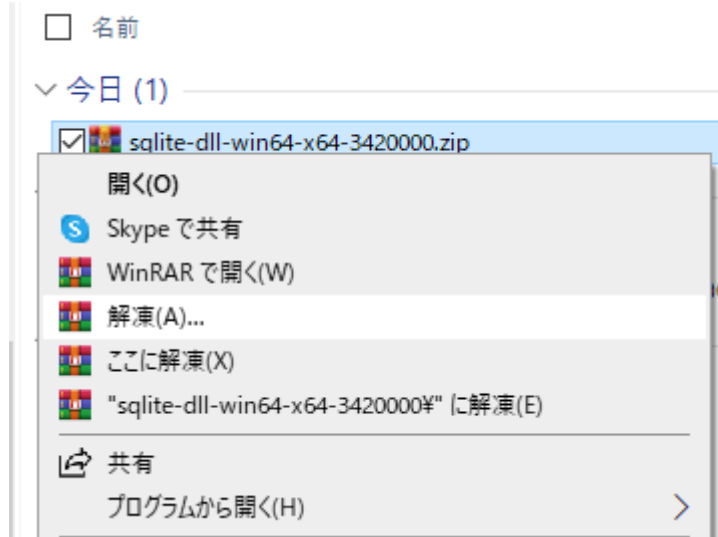
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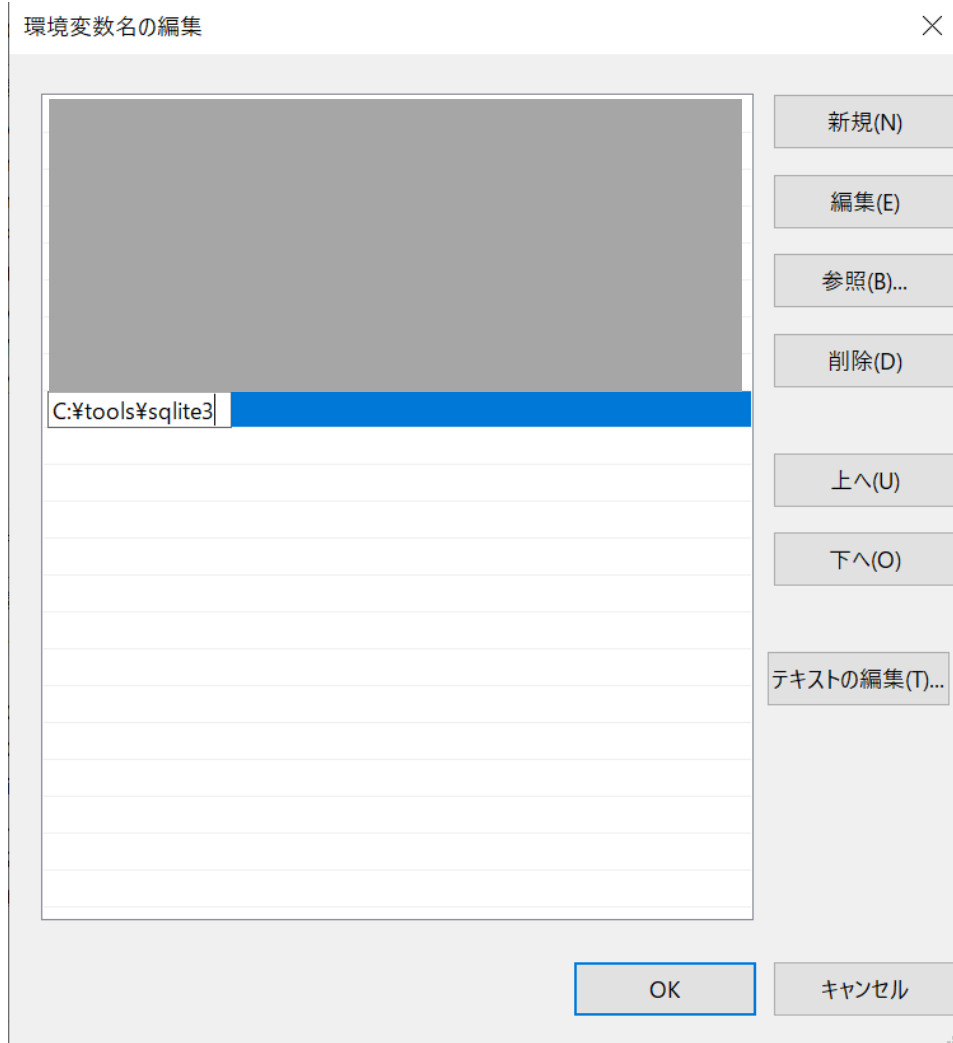
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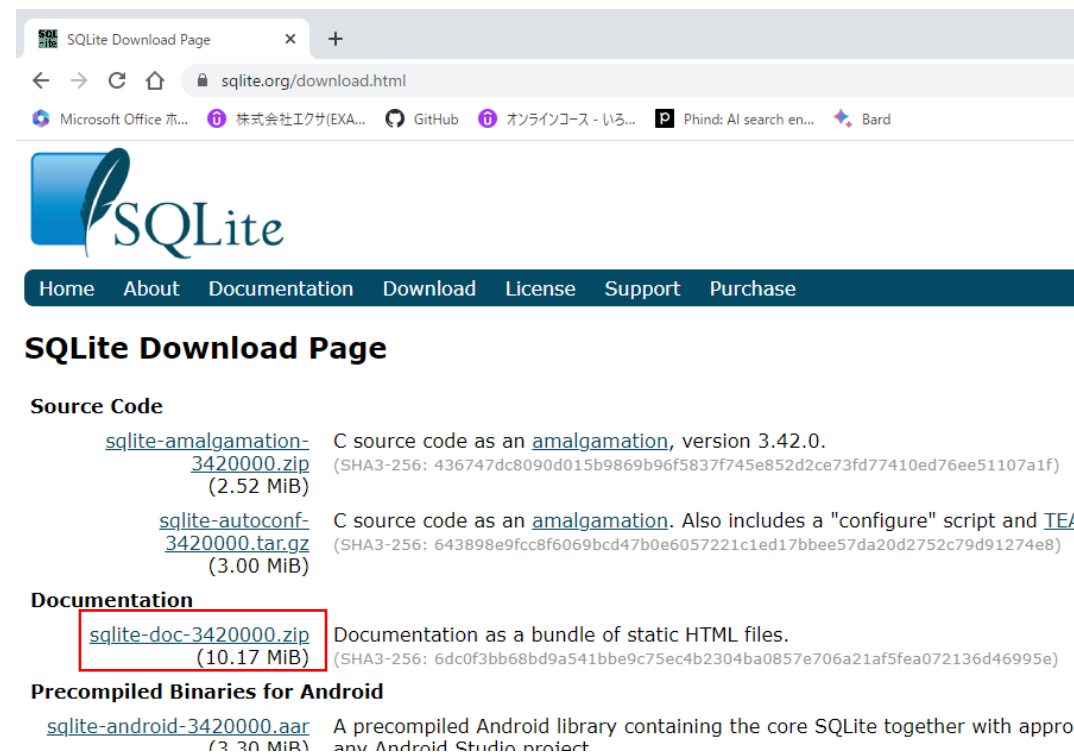
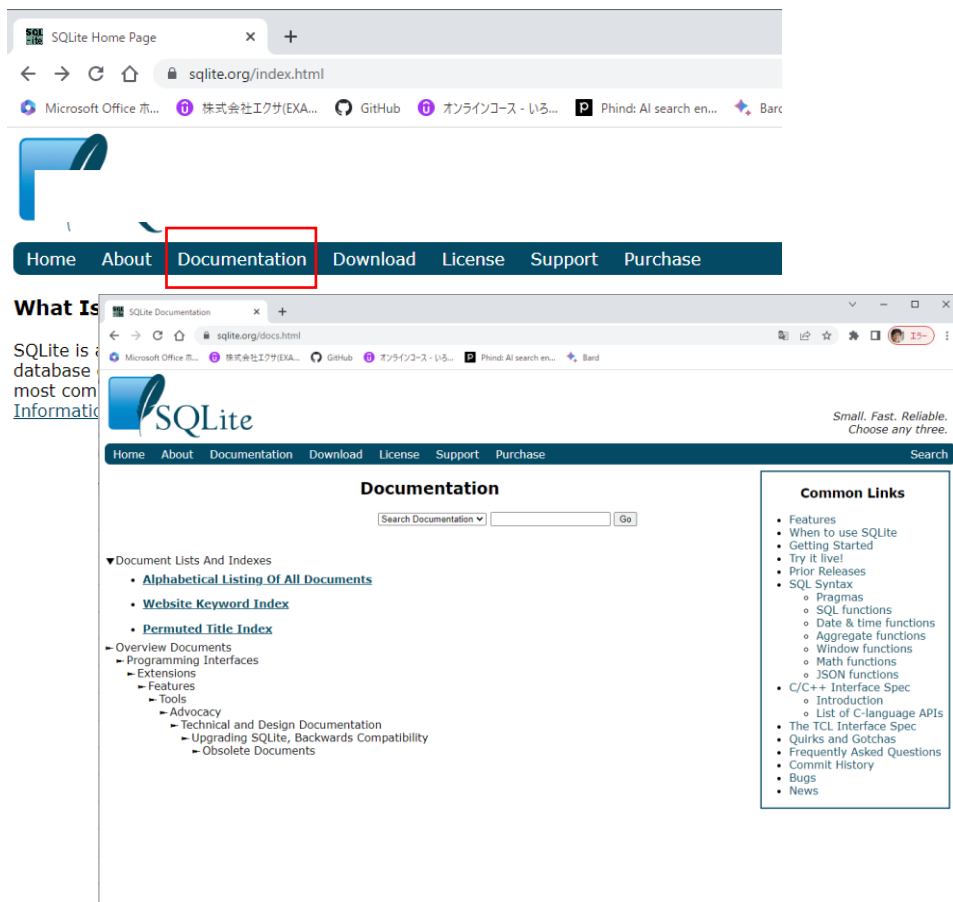
解凍ファイル(2つ)を同じフォルダへ格納



環境変数Pathに格納フォルダを追加



参考) SQLiteドキュメント



SQLite入門(日本語) : <https://www.javadrive.jp/sqlite/>

データベース作成

```
C:\Windows\system32\cmd.exe - sqlite3 booklist.sqlite3

(base) C:\tmp>sqlite3 booklist.sqlite3
SQLite version 3.42.0 2023-05-16 12:36:15
Enter ".help" for usage hints.
sqlite>
```



CSVファイルのインポート

```
sqlite> .mode csv
sqlite> .import data/orders.csv orders
sqlite> .import data/books.csv books
sqlite> .import data/employees.csv employees
sqlite>
```



インデックスの作成

```
sqlite> create index ordersindex on orders(id, branch);  
sqlite> create index booksindex on books(isbn);  
sqlite> create index employeesindex on employees(emp_no);  
sqlite>
```



スキーマの確認

```
sqlite> .schema orders
CREATE TABLE IF NOT EXISTS "orders"(
  "id" TEXT, "branch" TEXT, "isbn" TEXT, "recorddate" TEXT,
  "emp_no" TEXT, "buy" TEXT, "price" TEXT);
CREATE INDEX ordersindex on orders(id, branch);
sqlite> .schema books
CREATE TABLE IF NOT EXISTS "books"(
  "isbn" TEXT, "title" TEXT, "subtitle" TEXT, "writer" TEXT,
  "print" TEXT);
CREATE INDEX booksindex on books(isbn);
sqlite> .schema employees
CREATE TABLE IF NOT EXISTS "employees"(
  "emp_no" TEXT, "post" TEXT, "name" TEXT, "mail" TEXT);
CREATE INDEX employeesindex on employees(emp_no);
sqlite>
```



各テーブルの件数確認

```
sqlite> select count(*) from orders;
283
sqlite> select count(*) from books;
241
sqlite> select count(*) from employees;
95
sqlite> select count(*) from orders, books, employees where orders.isbn = books.isbn and orders.emp_no = employees.emp_no;
283
sqlite>
```



データベース ER図

サンプル：booklist.sqlite3

【書籍マスタ】 241件

books		
isbn	TEXT	ISBNコード
title	TEXT	書籍主題
subtitle	TEXT	書籍副題
writer	TEXT	著者
print	TEXT	出版元

【購入トランザクション】 283件

orders			
id	TEXT	複合キー	
branch	TEXT	複合キー	
isbn	TEXT	ISBNコード	
recorddate	TEXT	購入日	
emp_no	TEXT	購入依頼者職番	
buy	TEXT	購入元	
price	TEXT	購入価格	

【従業員マスタ】 95件

employees		
emp_no	TEXT	購入依頼者職番
post	TEXT	購入依頼者所属名
mail	TEXT	購入依頼者メール
name	TEXT	購入依頼者氏名



Sqlite3 コマンドの終了

```
sqlite> .exit  
(base) C:\tmp>
```

- 仕様上、Windows10で作成したSqlite3データベースファイルをLinuxOSなどにコピーしても使用可能
- 文字化けなどが発生した場合はCSVファイルを使って再作成のこと



SQLite へのアクセス(node)

- npm install sqlite3

《node》

```
const sqlite3 = require("sqlite3");  
const db = new sqlite3.Database("./data/booklist.sqlite3");
```



SQLiteへのアクセス(Java)

- [SQLite3 JDBCドライバ](#)をダウンロード
- CLASSPATHに追加

```
import java.sql.Connection;  
import java.sql.DriverManager;  
:  
Connection conn = DriverManager.getConnection(  
    "jdbc:sqlite:C:¥¥tmp¥¥data¥¥booklist.sqlite3");
```



SQLiteへのアクセス(Python)

※pip不要

```
import sqlite3
```

```
:
```

```
conn = sqlite3.connect("C:/tmp/data/booklist.sqlite3")
```



SQLite アンインストール手順

- 環境変数Pathに追加したパスを削除する
- SQLite フォルダを削除する

